Homework 1

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```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

#1.Loading tlc dataset

```
#install.packages("haven")
library(haven)

tlc <- read_sas("tlc.sas7bdat")
```

#2. Select just the placebo group (trt = P)

```
just_placebo <- tlc %>%
  subset(trt == "P")
```

#3. Find mean and variance of Pb (lead) at y0, y1, y4, and y6

```
#I could write a function and lapply through this, but this is simple and I'm lazy

y0_mean <- mean(just_placebo$y0)
y0_var <- var(just_placebo$y0)
y0_mean
```

```
## [1] 26.272
```

```r
y0_var
```

```
## [1] 25.24165
```

```r
y1_mean <- mean(just_placebo$y1)
y1_var <- var(just_placebo$y1)
y1_mean
```

```
## [1] 24.66
```

```r
y1_var
```

```
## [1] 29.82449
```

```r
y4_mean <- mean(just_placebo$y4)
y4_var <- var(just_placebo$y4)
y4_mean
```

```
## [1] 24.07
```

```r
y4_var
```

```
## [1] 33.09847
```

```r
y6_mean <- mean(just_placebo$y6)
y6_var <- var(just_placebo$y6)
y6_mean
```

```
## [1] 23.646
```

```r
y6_var
```

```
## [1] 31.80743
```

#4. Find covariance matrix of y0, y1, y4, and y6

```r
#selecting just the columns we're interested in
cov_data <- just_placebo %>%
  select(y0, y1, y4, y6)

#getting the covariances
cov_matrix <- cov(cov_data)
cov_matrix
```

```
##          y0       y1       y4       y6
## y0 25.24165 22.74947 24.26098 21.41784
## y1 22.74947 29.82449 27.04122 23.38412
## y4 24.26098 27.04122 33.09847 28.21896
## y6 21.41784 23.38412 28.21896 31.80743
```

#5. Confidence interval for the difference of mu_1 - mu_0 from estimated means from step 3 ##a. Assuming covariance is 0

```
#By-hand calculation formatted in R
y0_mean
```

```
## [1] 26.272
```

```
y1_mean
```

```
## [1] 24.66
```

```
y0_var
```

```
## [1] 25.24165
```

```
y1_var
```

```
## [1] 29.82449
```

```
n <- 50

l <- (24.66 - 26.272) - 1.98*sqrt((29.82 + 25.24)/50)
l
```

```
## [1] -3.689774
```

```
u <-(24.66 - 26.272) + 1.98*sqrt((29.82 + 25.24)/50)
u
```

```
## [1] 0.4657739
```

```
#these are the calculations, but I've put them in latex formatting for style too
```

Assuming a covariance of 0: $CI = (\bar{y}_1 - \bar{y}_0) \pm t_{\alpha/2,df} \times \sqrt{\frac{Var(y_1)+Var(y_0)}{n}}$

$CI = (24.66 - 26.272) \pm 1.98 \times \sqrt{\frac{29.82+25.24}{50}}$

Lower bound: $CI = (24.66 - 26.272) - 1.98 \times \sqrt{\frac{29.82+25.24}{50}}$ -3.6898

Upper bound: $CI = (24.66 - 26.272) + 1.98 \times \sqrt{\frac{29.82+25.24}{50}}$ 0.4658

Confidence interval: {-3.6898, 0.4658}

##b. Using covariance from step 4 for mu_1 and mu_0

```
#just pulling the covariance we need here.
cov_y1_y0 <- cov_matrix["y0", "y1"]
cov_y1_y0
```

```
## [1] 22.74947
```

```r
v <- (24.66 - 26.272) - 1.98*sqrt(((29.82 + 25.24) - 2*(22.750))/50)
v
```

```
## [1] -2.477783
```

```r
m <- (24.66 - 26.272) + 1.98*sqrt(((29.82 + 25.24) - 2*(22.750))/50)
m
```

```
## [1] -0.7462168
```

```r
#same as above, these are the calculations done "by hand", formatted below
```

With this covariance: $CI = (\bar{y}_1 - \bar{y}_0) \pm t_{\alpha/2,df} \times \sqrt{\frac{Var(y_1)+Var(y_0)-2Cov(y_0,y_1)}{n}}$

$CI = (24.66 - 26.272) \pm 1.98 \times \sqrt{\frac{29.82+25.24-2(22.750)}{50}}$

Lower bound: $CI = (24.66 - 26.272) - 1.98 \times \sqrt{\frac{29.82+25.24-2(22.750)}{50}}$ -2.478

Upper bound: $CI = (24.66 - 26.272) + 1.98 \times \sqrt{\frac{29.82+25.24-2(22.750)}{50}}$ -0.746

Confidence interval: {-2.478, -0.746}

#6. Two-sample t test comparing y1 and y0

```r
t.test(just_placebo$y1, just_placebo$y0, var.equal = TRUE)
```

```
##
##  Two Sample t-test
##
## data:  just_placebo$y1 and just_placebo$y0
## t = -1.5361, df = 98, p-value = 0.1277
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -3.6945781  0.4705781
## sample estimates:
## mean of x mean of y
##    24.660    26.272
```

CI here is {-3.695, 0.471}

#7. Paired t test of the change from week 0 to week 1 calculated in step 7

```r
t.test(just_placebo$y1, just_placebo$y0, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  just_placebo$y1 and just_placebo$y0
## t = -3.6852, df = 49, p-value = 0.0005711
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
```

```
##  -2.4910462 -0.7329538
## sample estimates:
## mean difference
##          -1.612
```

CI here is {-0.733, -2.491}

#8. Compare CI

A: R's means, by hand, covariance = 0 (-3.6896, 0.4658)

B: R's means, by hand, covariance from step 4 (-2.478, -0.746)

C: SAS's means, by hand, covariance = 0 (-3.6898, 0.4658)

D: SAS's means, by hand, covariance from step 4 (-2.478, -0.746)

E: From R's two-sample t test, comparing y1 with y0 (-3.695, 0.4706)

F: From R's paired t test of change from week 0 to week 1 (-2.491, -0.733)

G: From SAS's two-sample t test, comparing y1 with y0 (-0.4706, 3.6948)

H: From SAS's paired t test of change from week 0 to week 1 (-2.491, -0.733)

Assuming covariance = 0 and doing a two-sample t test seem to result in the same output of confidence intervals, regardless of where the means come from or in which program the test is run; A, C, E, and G should all be effectively the same, though obviously there has been some error in the calculation of G. The data processing in SAS of G results in the signs being flipped between the upper and lower bounds.

Utilizing the covariance from step 4 in both SAS and R results in the same CIs as well; B, D, F, and H are all as similar to each other as the previous group was. They tend to skew a little smaller than the covariance = 0 group, though, with the bounds staying below 0 while the two-sample and cov = 0 group is wider and bridges 0.